05 – Explainable AI (Phase 8 and the app's explanations)

SHAP explanations of the chosen networks, grouped by question group (H7), and a live explanation of one made-up example student exactly as the app produces it.

In [1]:
```
# Setup: work from the project root so the project code and configs are found
import os, sys, json, platform, warnings
from pathlib import Path
ROOT = Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd()
os.chdir(ROOT); sys.path.insert(0, str(ROOT))
warnings.filterwarnings("ignore")
import numpy as np, pandas as pd
from IPython.display import Image, Markdown, display
pd.set_option("display.precision", 3); pd.set_option("display.width", 160);
pd.set_option("display.max_columns", 20)
from src.utils.config import load_config, resolve_path
REPORTS = resolve_path(load_config("project")["paths"]["reports"])
FIG = resolve_path(load_config("project")["paths"]["figures"])
print("Project folder:", ROOT)
print("Python", platform.python_version(), "| numpy", np.__version__, "|
pandas", pd.__version__)
```

```
Project folder: C:\AL_outcome_predictor
Python 3.12.0 | numpy 2.5.4 | pandas 3.0.6
```

1. Which question groups matter (H7, permutation importance on validation rows)

In [2]:
```
pd.read_csv(REPORTS / "phase08" / "h7_group_importance.csv")
```

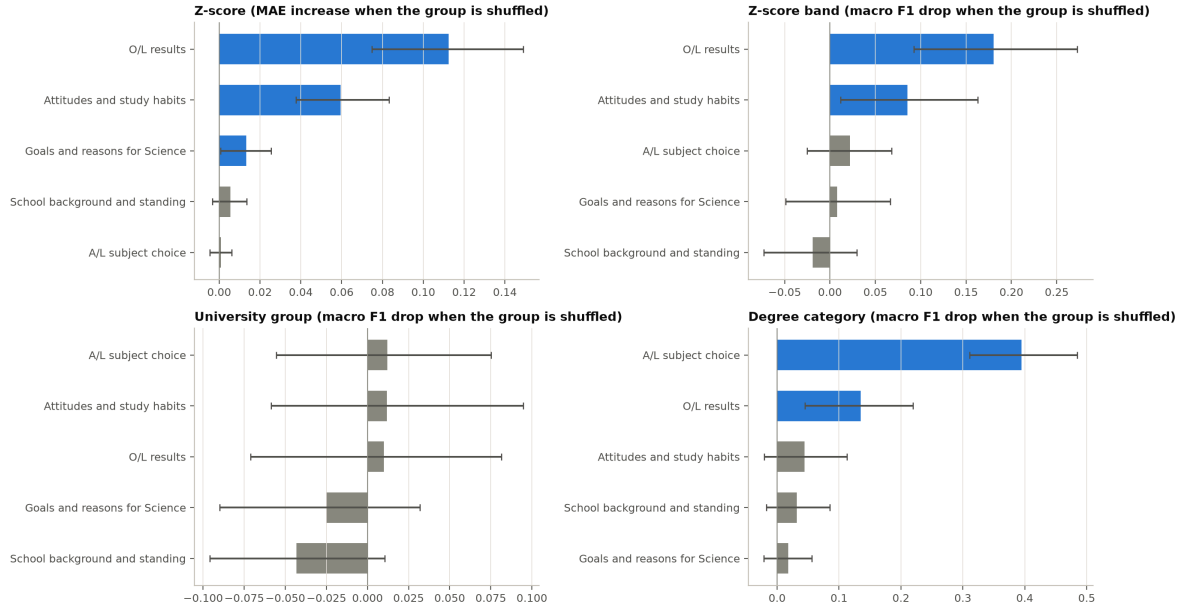
Out[2]:

	task	group	n_columns	importance	single_permutation	ci_low	ci_high	supporte
0	zscore	O/L results	10	1.123e-01	0.098	7.477e-02	0.149	Tru
1	zscore	School background and standing	8	5.550e-03	0.007	-3.242e-03	0.014	Fals
2	zscore	Attitudes and study habits	8	5.958e-02	0.053	3.770e-02	0.083	Tru
3	zscore	Goals and reasons for Science	4	1.322e-02	0.006	7.523e-04	0.026	Tru
4	zscore	A/L subject choice	4	9.511e-04	-0.002	-4.506e-03	0.006	Fals
5	zscore_band	O/L results	10	1.805e-01	0.129	9.264e-02	0.273	Tru
6	zscore_band	School background and standing	8	-1.914e-02	0.022	-7.281e-02	0.030	Fals
7	zscore_band	Attitudes and study habits	8	8.547e-02	0.105	1.162e-02	0.163	Tru
8	zscore_band	Goals and reasons for Science	4	8.003e-03	0.005	-4.864e-02	0.066	Fals
9	zscore_band	A/L subject choice	4	2.213e-02	0.030	-2.501e-02	0.068	Fals
10	university_group	O/L results	10	9.842e-03	-0.019	-7.126e-02	0.081	Fals
11	university_group	School background and standing	8	-4.315e-02	-0.023	-9.580e-02	0.010	Fals
12	university_group	Attitudes and study habits	8	1.177e-02	-0.008	-5.865e-02	0.095	Fals
13	university_group	Goals and reasons for Science	4	-2.531e-02	-0.022	-8.995e-02	0.032	Fals
14	university_group	A/L subject choice	4	1.209e-02	-0.016	-5.545e-02	0.075	Fals
15	degree_category	O/L results	10	1.349e-01	0.171	4.501e-02	0.219	Tru
16	degree_category	School background and standing	8	3.184e-02	0.067	-1.704e-02	0.085	Fals
17	degree_category	Attitudes and study habits	8	4.396e-02	0.030	-2.114e-02	0.113	Fals

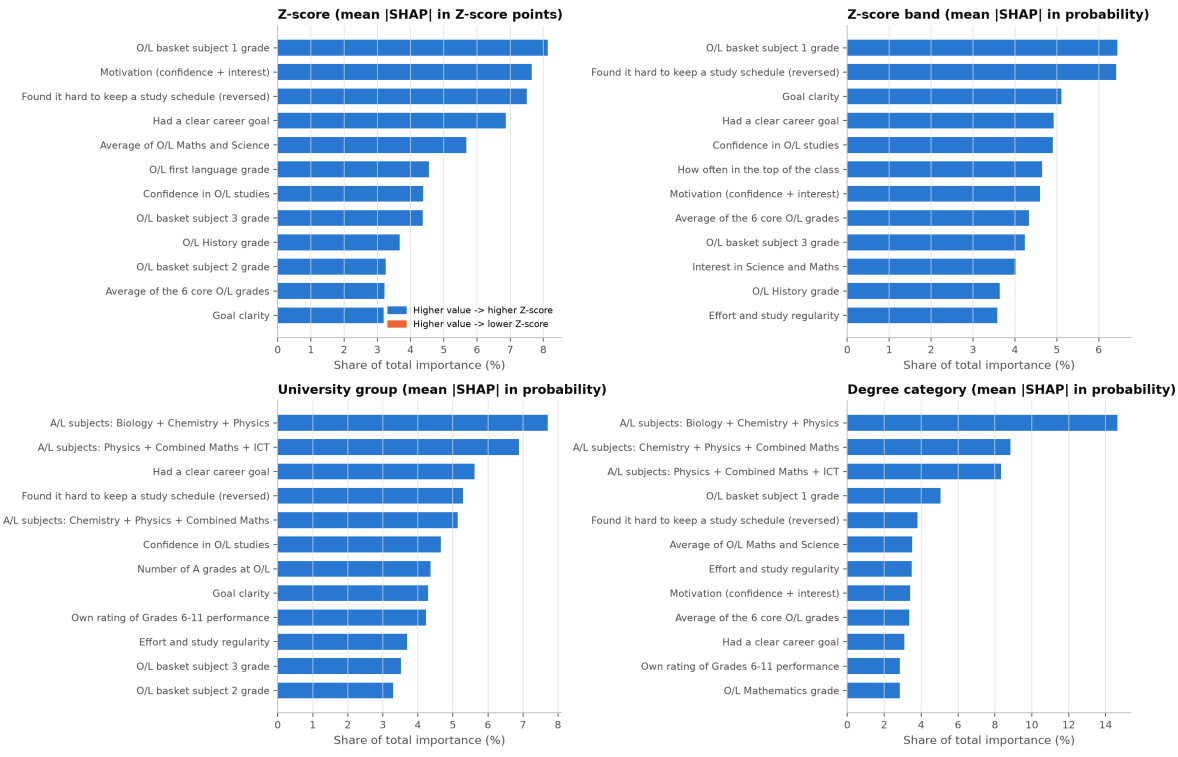
	task	group	n_columns	importance	single_permutation	ci_low	ci_high	supporte
18	degree_category	Goals and reasons for Science	4	1.818e-02	0.010	-2.128e-02	0.056	Fals
19	degree_category	A/L subject choice	4	3.944e-01	0.394	3.110e-01	0.485	Tru

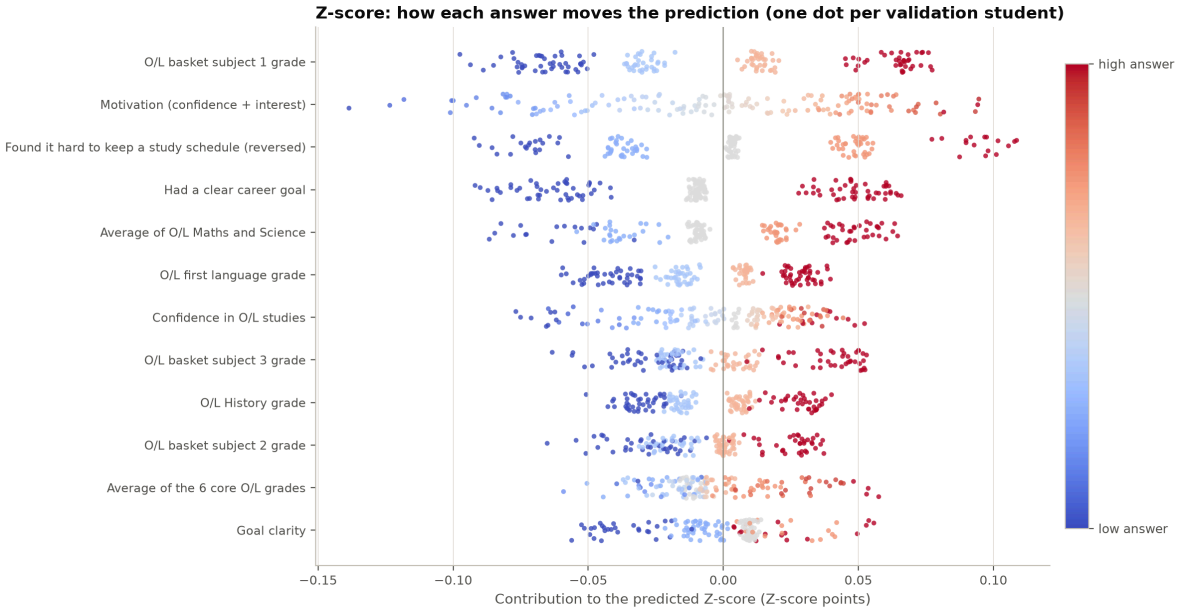
In [3]:
```
for name in ("phase08_h7_group_importance.png", "phase08_shap_importance.png",
"phase08_shap_zscore_beeswarm.png"):
    display(Image(str(FIG / name), width=760))
```

H7 - grouped permutation importance (validation, 95 % CI; blue = CI above 0)



What the chosen system relies on (SHAP, validation students; association, not cause)





2. Live explanation for the made-up "average" example student

The example students are written by hand (`configs/example_profiles.yaml`); they are not survey respondents.

In [4]:

```
import yaml
from src.models.service import Predictor
from src.explain.student import StudentExplainer
profile =
yaml.safe_load(resolve_path("configs/example_profiles.yaml").read_text("utf-8"))
["average"]
predictor = Predictor.load()
explainer = StudentExplainer.load(predictor)
z = explainer.explain(profile, "zscore")
print(f"Typical student: {z['typical_student_value']:.3f} | this student:
{z['prediction']:.3f}")
print(pd.Series(z["groups"]).round(3).rename("change in Z-score by
group").to_string())
print(f"Typical + groups = {z['typical_student_value'] +
sum(z['groups'].values()):.3f} (additivity error {z['additivity_error']:.1e})")
print("Stable over 3 runs:", z["stable"], "| top-5 Spearman:",
z["stability_spearman_top5"])
for sentence in z["sentences"]:
    print("-", sentence)
```

Typical student: 1.564 | this student: 1.419
O/L results   -0.151
Attitudes and study habits   0.037
A/L subject choice   -0.015
Goals and reasons for Science   -0.014
School background and standing   -0.003
Typical + groups = 1.419 (additivity error 0.0e+00)
Stable over 3 runs: True | top-5 Spearman: 0.9999999999999999
- Your O/L results lowered your predicted Z-score by about 0.15, compared with a typical student in our survey.
- Your attitudes and study habits raised your predicted Z-score by about 0.04, compared with a typical student in our survey.
- Your A/L subject choice lowered your predicted Z-score by about 0.02, compared with a typical student in our survey.
- Your goals and reasons for choosing Science lowered your predicted Z-score by about 0.01, compared with a typical student in our survey.